

MGSC 695

Advanced Topics in Management Science 1

Assignment #1

CNN-Based Image Classification: CIFAR-10 and CIFAR-100

By

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Introduction

This project presents an experimental study on CNN-based image classification using the CIFAR-10 and CIFAR-100 datasets. We explored CNN architecture design, data augmentation techniques, hyperparameter optimization, and transfer learning. The objective was to iteratively improve model performance through controlled experiments.

CNN Architecture and Design Decisions

Model Architecture

A custom Convolutional Neural Network (CNN) was designed for both CIFAR-10 and CIFAR-100 datasets. The key architectural choices include:

- Input Layer: 32×32 color images with 3 channels (RGB).
- Convolutional Layers: Four layers with increasing filter sizes: 32, 64, 128, 256.
- Batch Normalization: Applied after each convolution to stabilize learning.
- Activation Function: Tuned between ReLU, Leaky ReLU, and GELU.
- Pooling Layers: Max pooling used to reduce dimensionality.
- Fully Connected Layers: One hidden layer with 1024 neurons before classification.
- Dropout: Applied with 0.5 probability to prevent overfitting.
- Batch Size: 128, ensuring efficient training.

CNN Model Implementation

First Experiment: Initial Training and Tuning

The first phase involved training the CNN model from scratch on CIFAR-10 and fine-tuning on CIFAR-100 using multiple hyperparameters.

Hyperparameter Search

- Activation Functions: relu, leaky_relu, gelu
- Kernel Sizes: 3, 5
- Optimizers: adam, sgd

- Epochs: 50

Best Hyperparameters Observed for CIFAR-10

```
{  
    "activation": "leaky_relu",  
    "kernel_size": 5,  
    "optimizer": "sgd"  
}
```

Final CIFAR-10 Training Performance

Epoch [45/50], Train Loss: 0.1357, Val Loss: 0.3730, Val Acc: 88.63%

✓ Early stopping triggered.

Second Experiment: Further Hyperparameter Tuning

To improve model performance, additional hyperparameter tuning was performed with more hyperparameter combinations.

Newly Tested Hyperparameters

```
results = run_experiments(  
    dataset_name='CIFAR10',  
    activation_list=[nn.ReLU, nn.LeakyReLU],  
    optimizer_list=['Adam', 'SGD'],  
    dropout_rates=[0.3, 0.5],  
    learning_rates=[0.001, 0.01],  
    num_epochs=20  
)
```

Best Hyperparameters Observed for CIFAR-10

```
{  
  "activation": "ReLU",  
  "optimizer": "Adam",  
  "dropout": 0.5,  
  "learning_rate": 0.001  
}
```

Third Experiment: Extended Training with Early Stopping

Using the best hyperparameters, we trained the model for 50 epochs with early stopping.

Final CIFAR-10 Training Performance

Epoch [35/50] - Train Loss: 0.0346 - Val Loss: 0.3076 - Val Acc: 91.46%

Epoch [36/50] - Train Loss: 0.0349 - Val Loss: 0.3078 - Val Acc: 91.49%

Epoch [37/50] - Train Loss: 0.0344 - Val Loss: 0.2936 - Val Acc: 91.79%

✓ Early stopping triggered.

CIFAR-100 Transfer Learning Approach

After successfully training CIFAR-10, transfer learning was applied by initializing the CIFAR-100 model with the final trained CIFAR-10 weights. This approach allowed for faster convergence and improved accuracy.

Final CIFAR-100 Training Performance

Epoch [19/50] - Train Loss: 0.3348 - Val Loss: 1.2503 - Val Acc: 68.33%

Epoch [20/50] - Train Loss: 0.3176 - Val Loss: 1.2506 - Val Acc: 68.48%

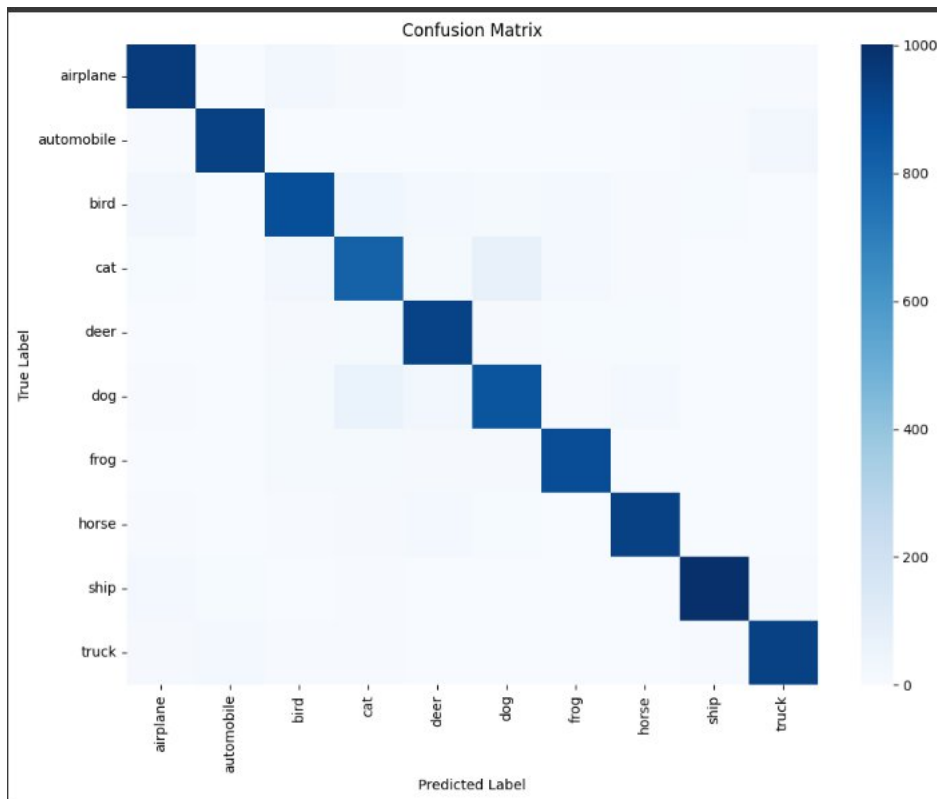
Epoch [21/50] - Train Loss: 0.3092 - Val Loss: 1.2521 - Val Acc: 68.23%

Epoch [22/50] - Train Loss: 0.3075 - Val Loss: 1.2654 - Val Acc: 68.02%

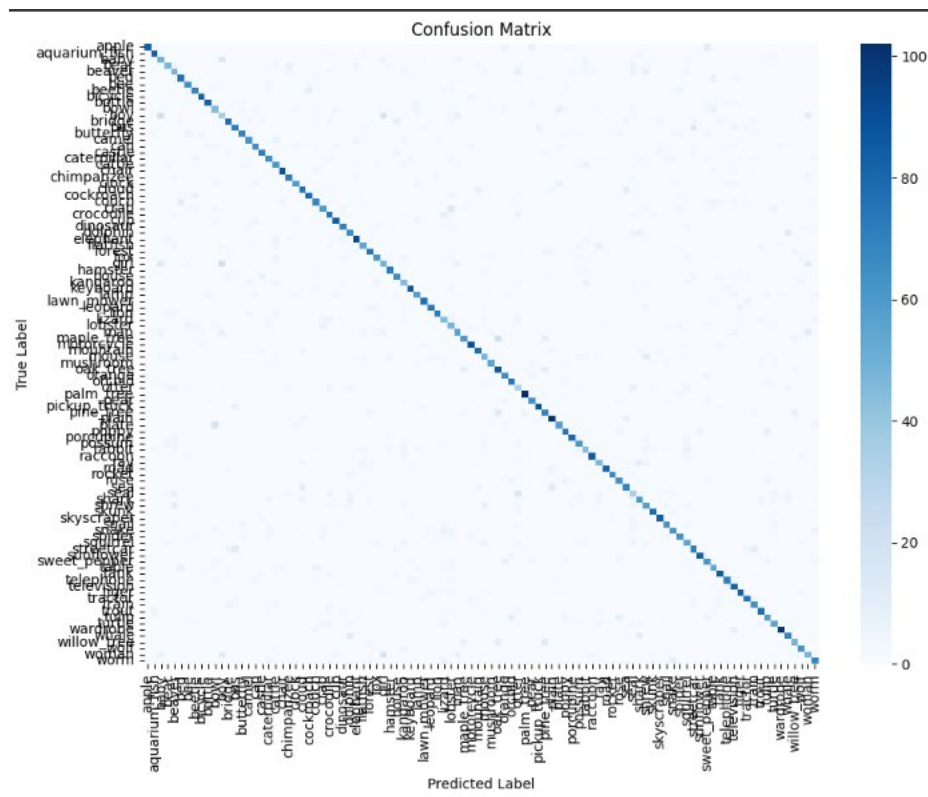
✓ Early stopping triggered.

Model Evaluation

CIFAR-10 Confusion Matrix



CIFAR-100 Confusion Matrix

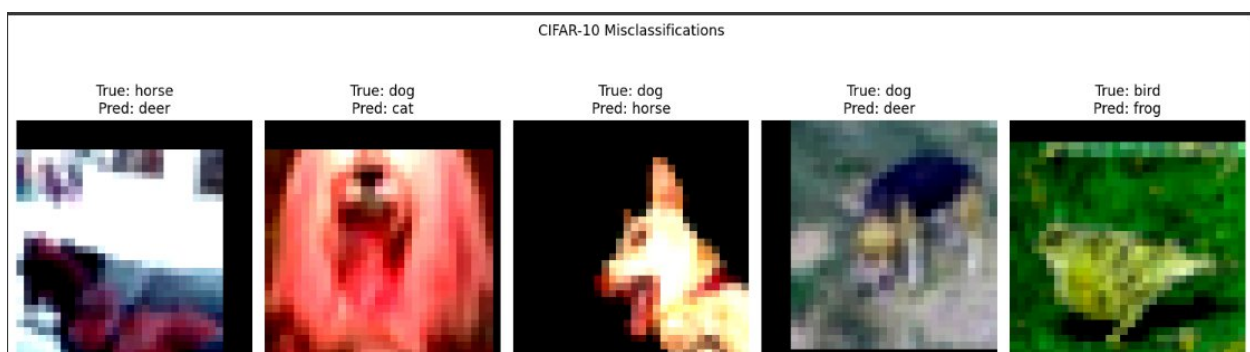


Insights:

- Transfer learning significantly improved CIFAR-100 accuracy compared to training from scratch.
- The model struggled with fine-grained categories, such as differentiating land animals from marine animals.

Misclassifications

CIFAR-10 Model



CIFAR – 100 model



Insights regarding potential causes of misclassification:

- * Faces of men and women can be visually similar in small 32x32 images with minimal resolution.
- * Both can share color palettes (bright orange, red), and low-res images can miss context.
- * Similar shape (quadruped posture), and texture ambiguity at low resolution.
- * Background confusion—both may appear in scenic, landscape-style images.

Ways to address them:

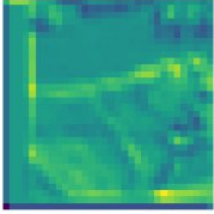
- * Use Transfer Learning with larger pre-trained models (like ResNet-50 on ImageNet).
- * Add hard example mining or focal loss to focus on tricky examples.
- * Use confusion matrix to drive curriculum learning: teach the model to resolve specific class confusion more carefully.

Feature Map Visualization and Grad-CAM

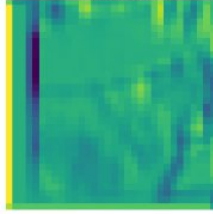
Feature Maps from Convolutional Layers

CIFAR-10 Model & CIFAR-100 Model

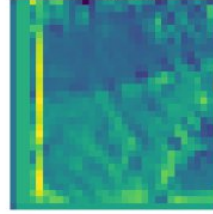
CIFAR-10 Model
conv1 | Ch 0



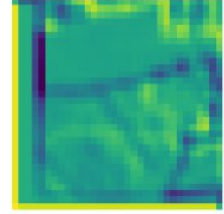
CIFAR-10 Model
conv1 | Ch 1



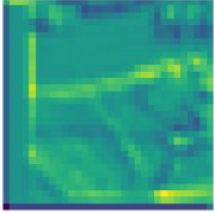
CIFAR-10 Model
conv1 | Ch 2



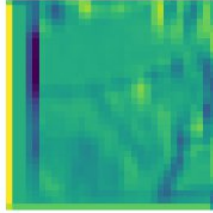
CIFAR-10 Model
conv1 | Ch 3



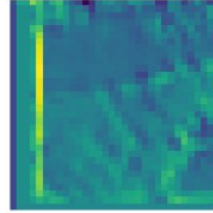
Fine-Tuned CIFAR-100 Model
conv1 | Ch 0



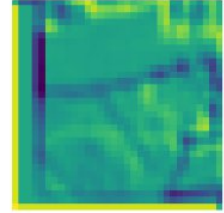
Fine-Tuned CIFAR-100 Model
conv1 | Ch 1



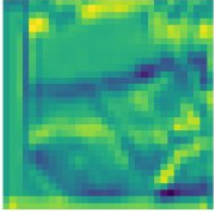
Fine-Tuned CIFAR-100 Model
conv1 | Ch 2



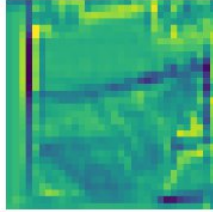
Fine-Tuned CIFAR-100 Model
conv1 | Ch 3



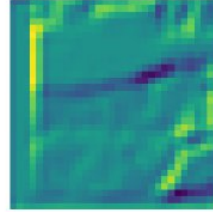
CIFAR-10 Model
res_block1.conv1 | Ch 0



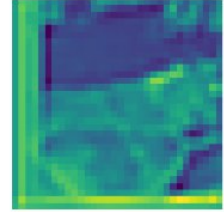
CIFAR-10 Model
res_block1.conv1 | Ch 1



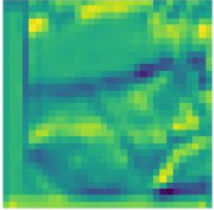
CIFAR-10 Model
res_block1.conv1 | Ch 2



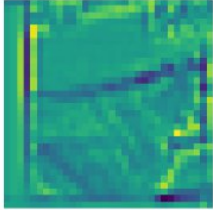
CIFAR-10 Model
res_block1.conv1 | Ch 3



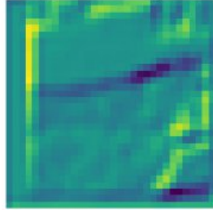
Fine-Tuned CIFAR-100 Model
res_block1.conv1 | Ch 0



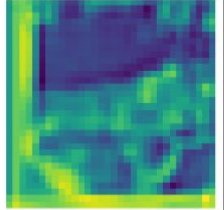
Fine-Tuned CIFAR-100 Model
res_block1.conv1 | Ch 1



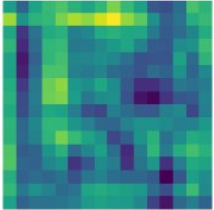
Fine-Tuned CIFAR-100 Model
res_block1.conv1 | Ch 2



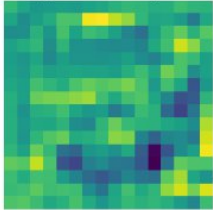
Fine-Tuned CIFAR-100 Model
res_block1.conv1 | Ch 3



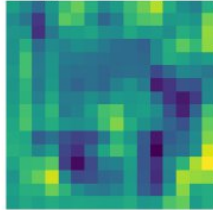
CIFAR-10 Model
res_block2.conv2 | Ch 0



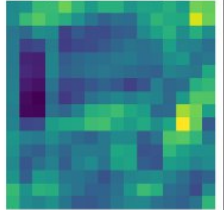
CIFAR-10 Model
res_block2.conv2 | Ch 1



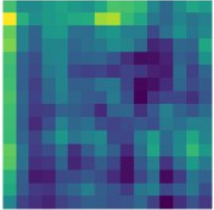
CIFAR-10 Model
res_block2.conv2 | Ch 2



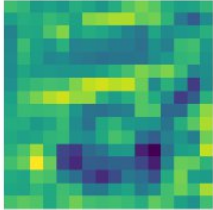
CIFAR-10 Model
res_block2.conv2 | Ch 3



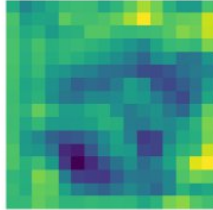
Fine-Tuned CIFAR-100 Model
res_block2.conv2 | Ch 0



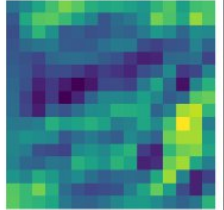
Fine-Tuned CIFAR-100 Model
res_block2.conv2 | Ch 1

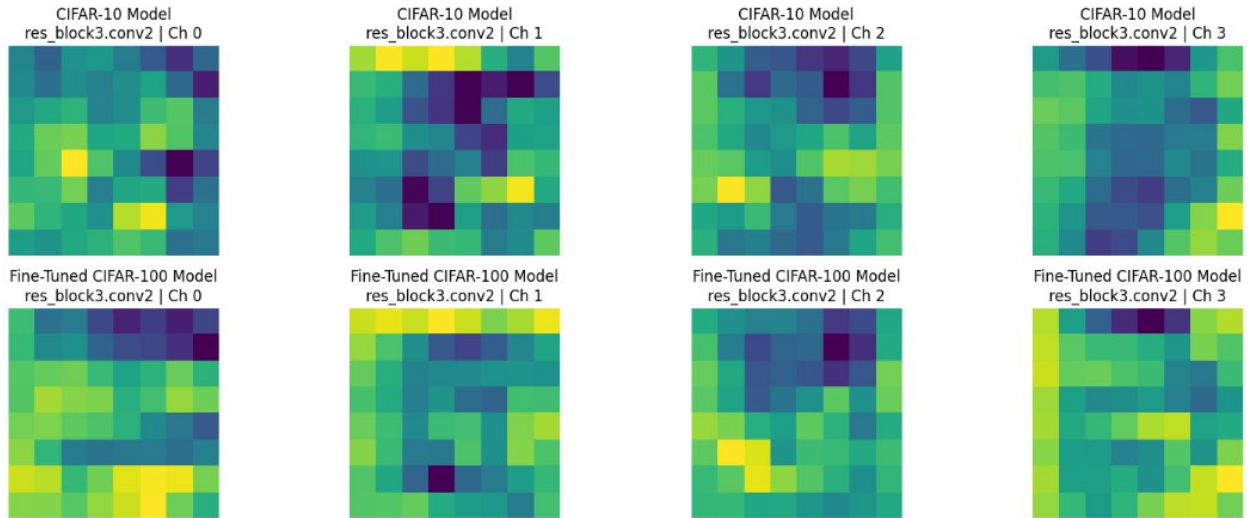


Fine-Tuned CIFAR-100 Model
res_block2.conv2 | Ch 2



Fine-Tuned CIFAR-100 Model
res_block2.conv2 | Ch 3



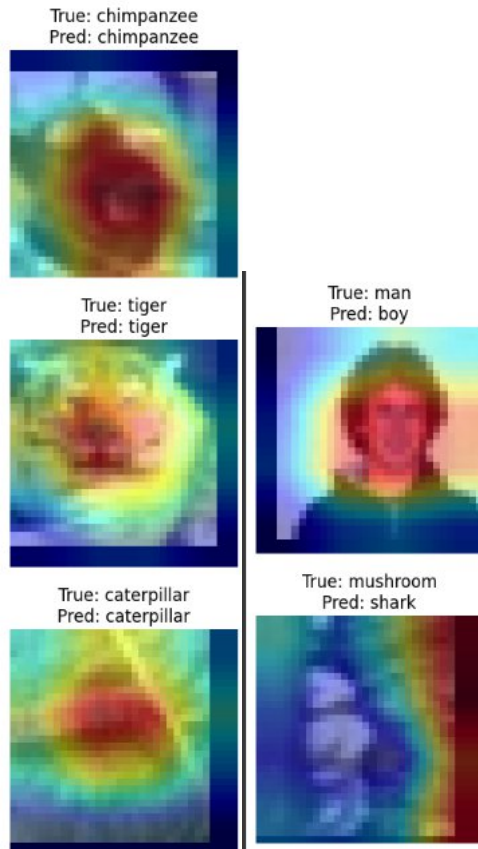


Grad-CAM Visualizations

CIFAR-10



CIFAR-100



Insights

We employed Grad-CAM to visualize feature attention of the CNN. Correctly classified samples (e.g., tiger, caterpillar) showed strong attention on object-centric regions, indicating robust feature learning. In contrast, misclassified cases (e.g., man predicted as boy) revealed diffuse or misleading attention, often influenced by background or overlapping features. These insights suggest potential gains from attention-focused training or hard negative mining.

Conclusion

- Best CIFAR-10 model achieved 91.79% accuracy after hyperparameter tuning and extended training.
- Best CIFAR-100 model reached 68.48% accuracy using transfer learning.
- Grad-CAM visualizations confirm that deeper layers focus on key object parts for classification.

- Future Work: Exploring ResNet, Vision Transformers (ViTs), and better augmentations to enhance performance.